



New Product Performance Study – MicroKlear 418AL

Introduction

MicroKlear 418 is one of Micro Powders' most popular additives for coatings requiring improved surface slip and scratch resistance with excellent clarity and gloss retention. MicroKlear 418AL is a micronized composite based on carnauba wax reinforced with 300 nm aluminum oxide nanoparticles (Mohs Hardness 9).

Property:	MKL-418	MKL-418AL
Mean particle size (µm)	6.0 – 8.0	6.0 – 8.0
Maximum particle size (µm)	22.0	22.0
Melting point (°C)	81 – 86	81 – 86
Density at 25 °C (g/cc)	0.99	1.04

Objective

Compare the level of scratch resistance between MicroKlear 418 and MicroKlear 418AL.

Study Methodology

Pencil scratch tests were conducted on a water based polyurethane coating:

Formula:	Weight %:
UROTUF® L522-W-40 (Reichhold)	89.6
Diethylene glycol monoethyl ether	5.0
BYK-345	0.5
TEGO® Foamex 805 N	0.3
Water	4.6
Total:	100.0
Nonvolatiles (Solids):	35.7

The polyurethane coating was dosed with 1% of MK 418 and MK 418AL. Coatings were drawn down with a #28 wire wound rod to produce approximately 1 mil DFT coatings. Coatings were dried for 7 days at room temperature and ~50% humidity.

The pencil scratch hardness test was conducted under controlled measurement conditions using a Taber Linear Abraser.



Taber Linear Abraser – Pencil Scratch Test Instrument

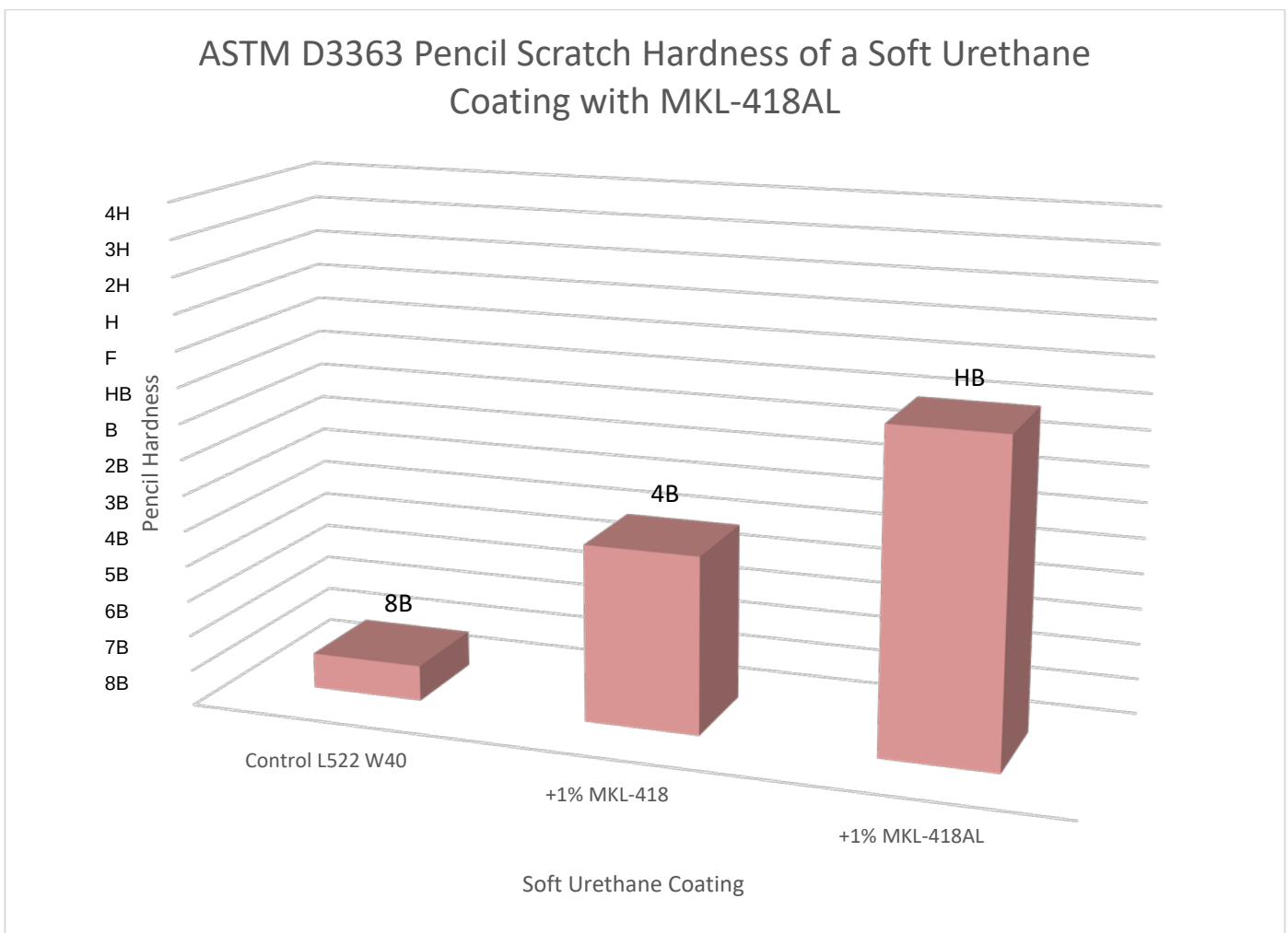
The motion of the abrasion is linear - one cycle is equal to one forward and one backward stroke of the test arm. The Taber Linear Abraser 5750, coupled with the Pencil Hardness kit accessory, enables controlled and reproducible Pencil Scratch Hardness testing of coatings by allowing the user to control the amount of force applied during the measurement, the number of linear cycles, the stroke length, as well as the speed. For this particular study, a force of 5.3 Newtons (equivalent to 536 g) was applied during the stroke. The stroke length was set to 0.5 inches and the speed was set to 2 cycles per minute. Each coating was scratched for one-half of a cycle (in the forward motion, as specified in ASTM D3363). These specific test conditions allowed one to observe differences in the amount of scratch for each dosed and undosed coating. Pencil scratch was assessed visually by comparing the surfaces of the scratched coatings and assigning the scratch hardness rating.



Results

Pencil Scratch Hardness Rating													
Soft	←					Mid-Point		→					Hard
8B	7B	6B	5B	4B	3B	2B	B	HB	F	H	2H	3H	4H

Test Samples	Pencil Scratch Hardness Rating
	1 MIL DFT
Control (No Additive)	8B
+ 1% MKL-418	4B
+ 1% MKL-418AL	HB



Conclusions

- MKL-418AL provides a dramatic improvement in Taber linear abrasion (pencil scratch hardness)
- MKL-418AL gives the formulator the cutting-edge performance of alumina nanopowder in a safe, easier-to-disperse, non-nano additive
- MKL-418AL has a density optimized for maximum surface efficiency and minimal settling vs. alumina nanopowders

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